The Nernst Equation and Cell Potential

Non-standard cells, concentration cells, E°, ΔG° and K

Name _____ Class _____ Date _____

EQUATIONS AND DATA (25 °C)

$E = E^\circ - (0.0592\ \text{V} / n) \log Q \cdot \Delta G^\circ = -nFE^\circ \cdot \log K = nE^\circ / 0.0592\ \text{V} \cdot F = 96\,485\ \text{C/mol}$

E° (V): Ag^+/Ag +0.80 · Cu^{2+}/Cu +0.34 · Zn^{2+}/Zn −0.76

$E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode}) \cdot 1\ \text{V} \times 1\ \text{C} = 1\ \text{J}$

Write the balanced overall equation first to find n and Q. Show units in every step.

A. Ideas

1. Explain why the voltage of a battery falls as it is used, and what the voltage is when it is completely flat.

B. Using the Nernst equation

2. For $Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$, calculate E when $[Zn^{2+}] = 0.10$ M and $[Cu^{2+}] = 2.0$ M. Is E larger or smaller than E°? Explain.

3. For $2\,Ag^+ + Cu \longrightarrow 2\,Ag + Cu^{2+}$: (a) calculate E° and give n (b) calculate E when $[Ag^+] = 0.010$ M and $[Cu^{2+}] = 1.0$ M.

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4. A concentration cell has silver electrodes in 0.0010 M and 0.10 M Ag^+. (a) Which electrode is the anode? (b) Calculate E.

C. E°, ΔG° and K

5. For the reaction in question 3, calculate (a) ΔG° (b) K at 25 °C.

6. A one-electron reaction has $K = 1.0 \times 10^5$ at 25 °C. Calculate E°.

7. Without calculating, what can you say about ΔG° and K for a cell with a negative E°?

D. Challenge

8. A Zn–Cu cell starts with E = 1.10 V. Calculate the ratio $[Zn^{2+}]/[Cu^{2+}]$ at which its voltage has fallen to 1.00 V.

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